

Parallel computing

Lab 08

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Exercise 0

```
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ mpirun --oversubscribe -np 2 ./HelloMPI
Just a normal process From rank 1 machine ip-172-31-33-227.ec2.internal
Hello World! From rank 0 machine ip-172-31-33-227.ec2.internal
```

Exercise 1

```
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ mpic++ -o messages1 messages1.cc
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ mpirun --oversubscribe -np 4 ./messages1
Just a normal process From rank 0 machine ip-172-31-33-227.ec2.internal
Just a normal process From rank 2 machine ip-172-31-33-227.ec2.internal
SEnding message to computer 3 from computer 1
in computer 3 the value of y is 30
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ |
```

Exercise 2

```
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ mpicc -o exercise2 exercise2.c
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ mpirun --oversubscribe -np 4 ./exercise2
Total sum = 5000000050000000
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ |
```

Exercise 3

```
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ mpicc -o exercise3 exercise3.c
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ mpirun --oversubscribe -np 4 ./exercise3
Estimated Pi = 3.1415384000
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ |
```

Exercise 4

```
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ mpicc -o exercise2 exercise2.c
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ mpicc -o exercise3 exercise3.c
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ time mpirun --oversubscribe -np 1 ./exercise2
Total sum = 50000005000000

real    0m1.518s
user    0m0.108s
sys     0m0.121s
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ time mpirun --oversubscribe -np 2 ./exercise2
Total sum = 50000005000000

real    0m1.678s
user    0m0.129s
sys     0m0.264s
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ time mpirun --oversubscribe -np 4 ./exercise2
Total sum = 50000005000000

real    0m2.016s
user    0m0.185s
sys     0m0.562s
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ time mpirun --oversubscribe -np 1 ./exercise3
Estimated Pi = 3.1405080000

real    0m1.942s
user    0m0.484s
sys     0m0.118s
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ time mpirun --oversubscribe -np 2 ./exercise3
Estimated Pi = 3.1419692000

real    0m2.067s
user    0m0.526s
sys     0m0.255s
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ time mpirun --oversubscribe -np 4 ./exercise3
Estimated Pi = 3.1413032000

real    0m2.406s
user    0m0.600s
sys     0m0.540s
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ |
```

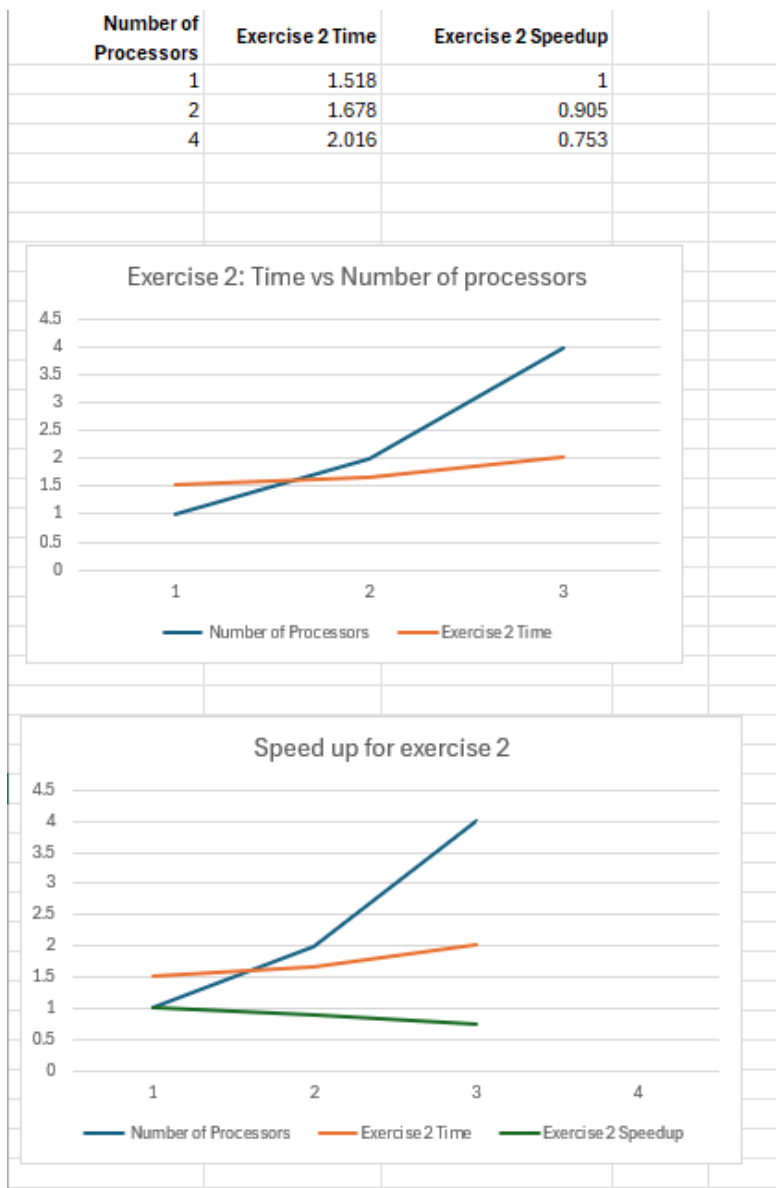
Exercise 2 results:

Number of Processors	Time (seconds)	Speedup
1	1.518	1.000
2	1.678	0.905
4	2.016	0.753

Exercise 3 results:

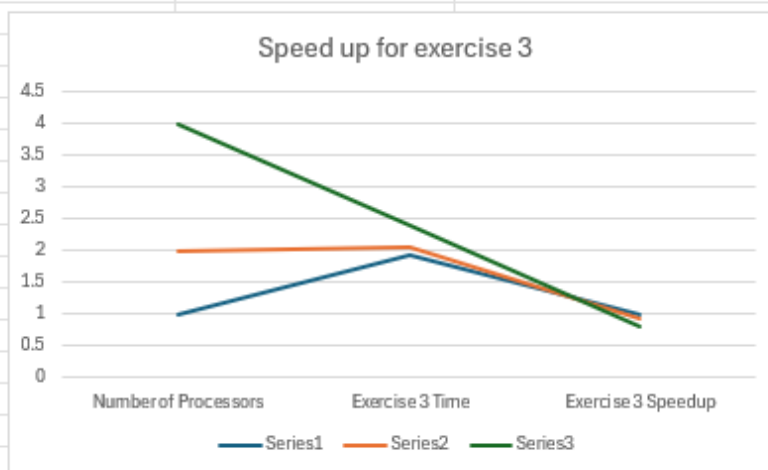
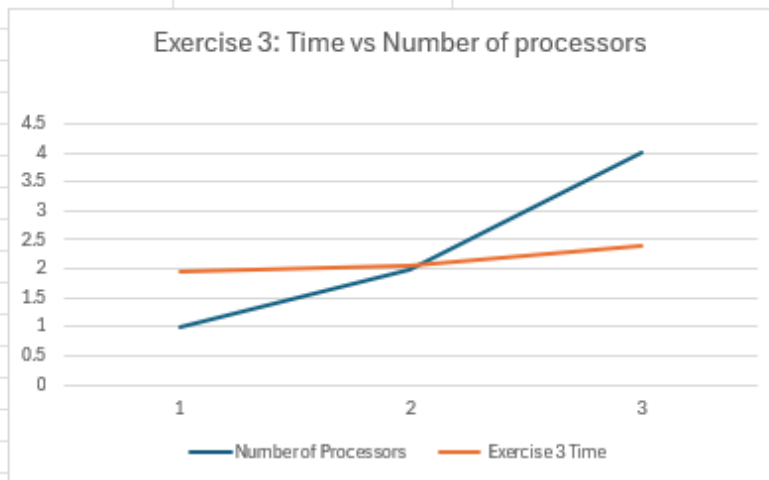
Number of Processors	Time (seconds)	Speedup
1	1.942	1.000
2	2.067	0.940
4	2.406	0.807

Plot for the exercise 2: time vs number of processors and speedup



Plot for the exercise 3: time vs number of processors and speed up

Number of Processors	Exercise 3 Time	Exercise 3 Speedup
1	1.942	1
2	2.067	0.94
4	2.406	0.807



Exercise 5

1.

```
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ cp message2.cc exercise5_1.cc
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ nano exercise5_1.cc
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ mpic++ -o exercise5_1 exercise5_1.cc
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ mpirun --oversubscribe -np 4 ./exercise5_1
Just a normal process From rank 0 machine ip-172-31-33-227.ec2.internal
Sending message to computer 3 from computer 1
Just a normal process From rank 2 machine ip-172-31-33-227.ec2.internal
```

When the source and destination do not match, the receiving process waits for a message from the specified source. In this case, Rank 3 waits for a message from Rank 2, but Rank 2 does not send any message. Therefore, the program becomes blocked and does not complete.

2.

```
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ mpic++ -o exercise5_2 exercise5_2.cc
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ mpirun --oversubscribe -np 4 ./exercise5_2
Just a normal process From rank 0 machine ip-172-31-33-227.ec2.internal
Sending buffered message to computer 3 from computer 1
In computer 3 the value of y is printed
 0 10 20 30 40 50 60 70 80 90
Just a normal process From rank 2 machine ip-172-31-33-227.ec2.internal
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ |
```

MPI_Bsend() was used to send the array from Rank 1 to Rank 3 using a user-attached buffer. The program executed successfully, and Rank 3 received and printed the values 0, 10, 20, ..., 90. This shows that buffered communication allows the message to be stored in the attached buffer before it is received by the destination process.

Exercise 6

```
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ nano exercise3.c
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ cp exercise3.c exercise6.c
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ nano exercise6.c
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ mpicc -o exercise6 exercise6.c
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ mpirun --oversubscribe -np 4 ./exercise6
Estimated Pi = 3.1412908000
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ |
```

In Exercise 3, `MPI_Reduce()` was used to combine the number of points inside the circle from all processes. In Exercise 6, the reduction was replaced with `MPI_Send()` and `MPI_Recv()` using `MPI_ANY_SOURCE`. The results are similar because both programs perform the same Monte Carlo calculation. However, the order in which messages are received in Exercise 6 can vary because Rank 0 accepts messages from any source.

Exercise 7

```
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ mpicc -o exercise7 exercise7.c
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ mpirun --oversubscribe -np 4 ./exercise7
Estimated Pi = 3.1420828000
[ec2-user@ip-172-31-33-227 PC_Lab_08]$ |
```